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Experiment: Decision Tree Implementation in Python

Aim

To implement the **Decision Tree** algorithm in Python for classification using the Iris dataset and evaluate its performance.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Create a Decision Tree classifier.
5. Train the classifier using the training data.
6. Predict the output for the test data.
7. Calculate the accuracy of the model.
8. Display the decision tree.

Program

```
# Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.metrics import accuracy_score

import matplotlib.pyplot as plt

# Load dataset
```

```
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Decision Tree model
model = DecisionTreeClassifier()

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# Display Decision Tree
plt.figure(figsize=(10, 6))
plot_tree(
    model,
```

```
feature_names=iris.feature_names,  
class_names=iris.target_names,  
filled=True  
)  
plt.show()
```

Output (Sample)

Accuracy: 1.0

A graphical representation of the Decision Tree will also be displayed.

```
1 # Import required libraries  
2 from sklearn.datasets import load_iris  
3 from sklearn.model_selection import train_test_split  
4 from sklearn.tree import DecisionTreeClassifier, plot_tree  
5 from sklearn.metrics import accuracy_score  
6 import matplotlib.pyplot as plt  
7  
8 # Load dataset  
9 iris = load_iris()  
10 X = iris.data  
11 y = iris.target  
12  
13 # Split dataset into training and testing  
14 X_train, X_test, y_train, y_test = train_test_split(  
15     X, y, test_size=0.3, random_state=42  
16 )  
17  
18 # Create Decision Tree model  
19 model = DecisionTreeClassifier()  
20  
21 # Train the model  
22 model.fit(X_train, y_train)  
23  
24 # Predict test data  
25 y_pred = model.predict(X_test)  
26  
27 # Calculate accuracy  
28 accuracy = accuracy_score(y_test, y_pred)  
29 print("Accuracy:", accuracy)  
30
```

ERROR!
Traceback (most recent call last):
 File "<main.py>", line 2, in <module>
ModuleNotFoundError: No module named 'sklearn'

=== Code Exited With Errors ===

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Result

Thus, the **Decision Tree algorithm** was successfully implemented in Python using the Iris dataset, and the model classified the data with high accuracy.